CSE634 - Algorithmic Techniques for Robot Planning (ARP)

Introduction

Instructor: Dr. Tanmoy Kundu

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Eligibility

- **Eligible students:** UG/PG
- **Prerequisites:**
  - Data Structures and Algorithms
  - Discrete Mathematics
- **Credits:** 4

Course description

- This course studies applications of algorithmic techniques – e.g. searching, greedy, divide-and-conquer, dynamic, graph algorithms – in robot path planning, task planning and their combinations.

- **Goal:** Introduce some advanced topics of algorithms and train to apply them in robotic applications.

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- Assignments will contain hands-on programming in *Robot Operating System (ROS) 2* and model checking tools (*PRISM, UPPAAL*).

Course description

- **Discrete planning**
  - Re-visiting known graph algorithms from planning perspective (Dijkstra's, A* algorithms)
  - Sampling based motion planning (PRM, RRT)

Course description

- **Model checking**
  - Reachability analysis
  - Linear Temporal Logic
  - Timed automata

Course description

- **Computational geometry**
  - Convex hull
  - Polygon triangulation
  - Voronoi diagrams

Course description

- **Planning under uncertainty**
  - Markov Decision Process (MDP)
  - Partially Observable Markov Decision Process (POMDP)
  - Dynamic programming
  - Monte Carlo methods

Assessment plan

- Assignments (10%)
- Quizzes (20%)
- Course project (15%)
- Midsem (25%)
- Endsem (30%)

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- You need not worry about novelty, applicability etc. I encourage you just to proceed with it.

Resources

- **Text books**
  - Planning Algorithms, Steven M. LaValle, Cambridge Press, 2006.
  - Principles of Model Checking, Christel Baier et. al., The MIT Press, 2008.
  - Computational Geometry, Mark de Berg et. al., Springer, 2008.
  - Reinforcement learning, Richard Sutton et. al., The MIT Press, 2018.
- **Reference books**
  - Introduction to Algorithms, Cormen et. al., The MIT Press, 2009.
  - Probabilistic robotics, Sebastian Thrun et. al., The MIT Press, 2005.
  - Research papers

Course description

- **Students registered: 142**
  - ECE: 1 (1 PhD)
  - CSSS: 3
  - CSB: 6
  - CSD: 8
  - CSAM: 16
  - CSAI: 39
  - CSE: 69 (including 2 PhDs and 10 M.Techs)

Thank you